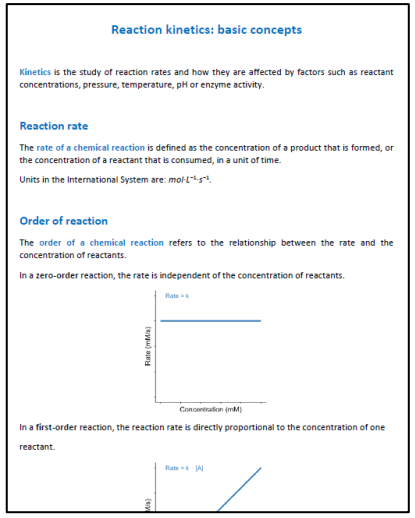
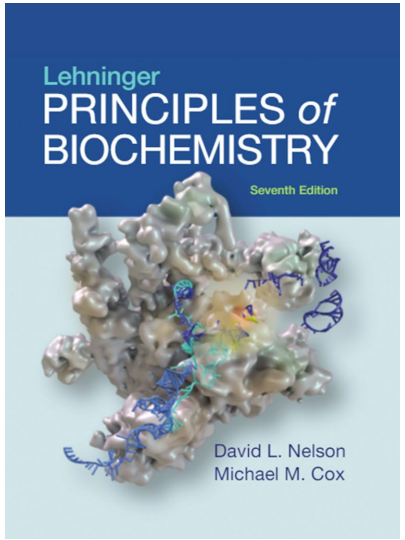


Enzymes

Biological catalysts: structure, function and regulation

Supportive material





Additional study material

Aula Virtual:

- Reaction kinetics

Reading

Lehninger's PoB Chapter 6

6.1. Introduction
6.2. How enzymes work
6.5. Regulatory enzymes

Learning goals

- Physiological significance of enzymes
- Enzyme characteristics as biocatalysts
- Characteristics of the binding site
- Concept of cofactor
- How enzymes work
- Mechanisms of enzyme regulation

What is an enzyme?

- Enzymes are catalysts

  They increase reaction rates without being used up

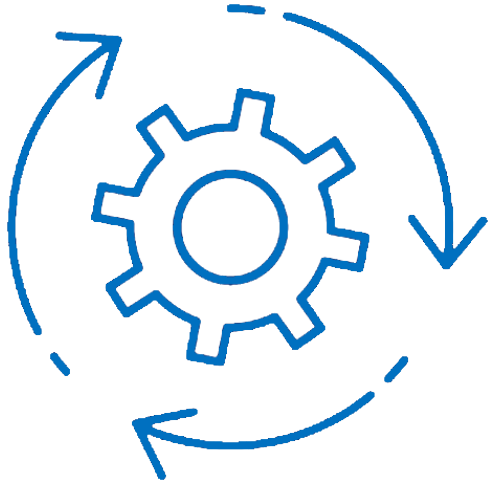
- Most enzymes are globular proteins

  However, some RNA molecules (ribozymes) also catalyze reactions

- Involved in almost all biological processes

$$\text{Sucrose } (C_{12}H_{22}O_{11}) + 12\ O_2 \rightarrow 12\ CO_2 + 11\ H_2O$$

Why enzymes rather than inorganic catalysts?



- Greater reaction specificity
  - Fewer side products
- Milder reaction conditions
  - Compatible with conditions in cells
- Higher catalytic power
  - In a biologically useful time scale
- Capacity for regulation
  - Control of biological pathways

Rate enhancement by enzymes

TABLE 6-5	Some Rate Enhancements Produced by Enzymes
Cyclophilin	10^5
Carbonic anhydrase	10^7
Triose phosphate isomerase	10^9
Carboxypeptidase A	10^{11}
Phosphoglucomutase	10^{12}
Succinyl-CoA transferase	10^{13}
Urease	10^{14}
Orotidine monophosphate decarboxylase	10^{17}

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Enzymes make life possible

General properties of enzymes

- Globular proteins
- Varied sizes: 10,000 - 10^6 Da
- Monomeric or multimeric
- Simple or conjugated
- Specific regions:
  - Active site (binding and catalysis)
  - Some enzymes also contain regulatory sites

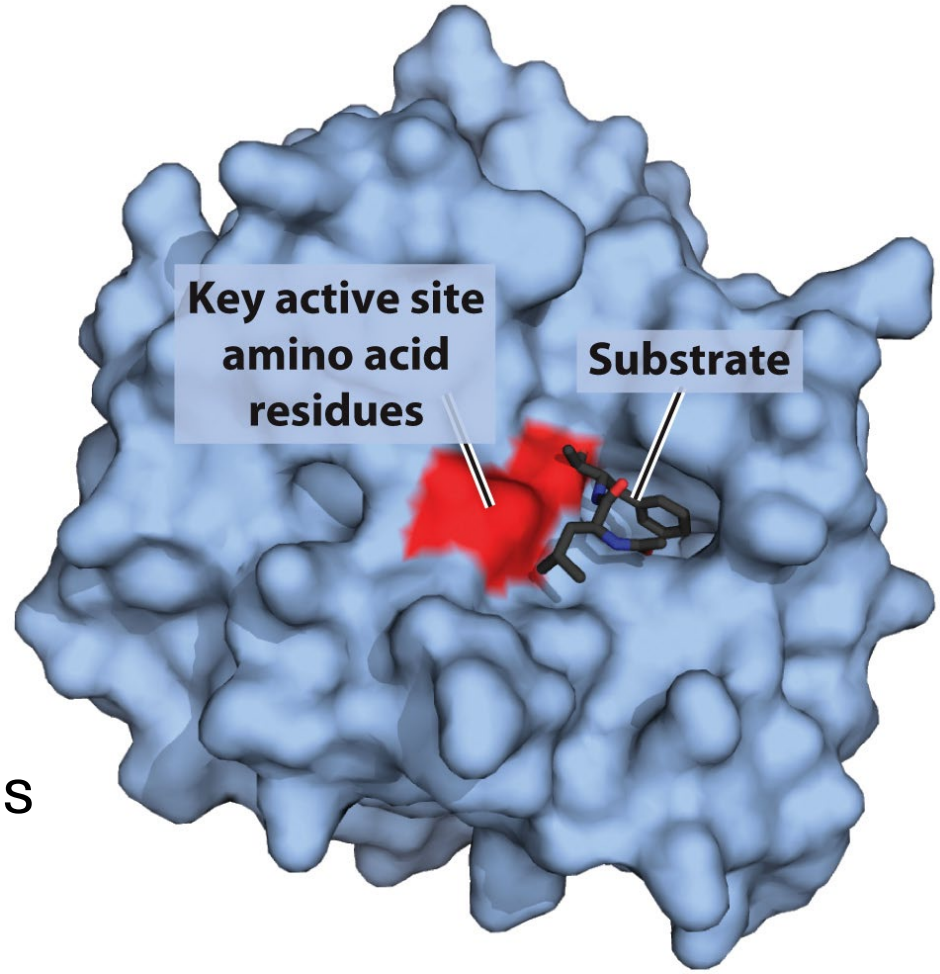


Figure 6-1
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Enzyme cofactors

Holoenzyme = apoenzyme + cofactor

Types of cofactors:

- Metal ions
- Organic molecules (coenzymes):
  - Prosthetic groups
  - Cosubstrates

Metal ions

TABLE 6-1	Some Inorganic Ions That Serve as Cofactors for Enzymes
Ions	**Enzymes**
Cu^{2+}	Cytochrome oxidase
Fe^{2+} or Fe^{3+}	Cytochrome oxidase, catalase, peroxidase
K^{+}	Pyruvate kinase
Mg^{2+}	Hexokinase, glucose 6-phosphatase, pyruvate kinase
Mn^{2+}	Arginase, ribonucleotide reductase
Mo	Dinitrogenase
Ni^{2+}	Urease
Zn^{2+}	Carbonic anhydrase, alcohol dehydrogenase, carboxypeptidases A and B

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Coenzymes

TABLE 6-2	Some Coenzymes That Serve as Transient Carriers of Specific Atoms or Functional Groups	
Coenzyme	**Examples of chemical groups transferred**	**Dietary precursor in mammals**
Biocytin	CO_2	Biotin
Coenzyme A	Acyl groups	Pantothenic acid and other compounds
5'-Deoxyadenosylcobalamin (coenzyme B_{12})	H atoms and alkyl groups	Vitamin B_{12}
Flavin adenine dinucleotide	Electrons	Riboflavin (vitamin B_2)
Lipoate	Electrons and acyl groups	Not required in diet
Nicotinamide adenine dinucleotide	Hydride ion ($:H^-$)	Nicotinic acid (niacin)
Pyridoxal phosphate	Amino groups	Pyridoxine (vitamin B_6)
Tetrahydrofolate	One-carbon groups	Folate
Thiamine pyrophosphate	Aldehydes	Thiamine (vitamin B_1)

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Nomenclature

Diverse identification systems

- Name of the substrate + *-ase*:

  urease, lactase…

- Substrate + enzymatic activity ending with *-ase*:

  triose phosphate isomerase, DNA polymerase…

- Common name:

  trypsin, rubisco, papain…

How enzymes work

$S \leftrightarrow P$

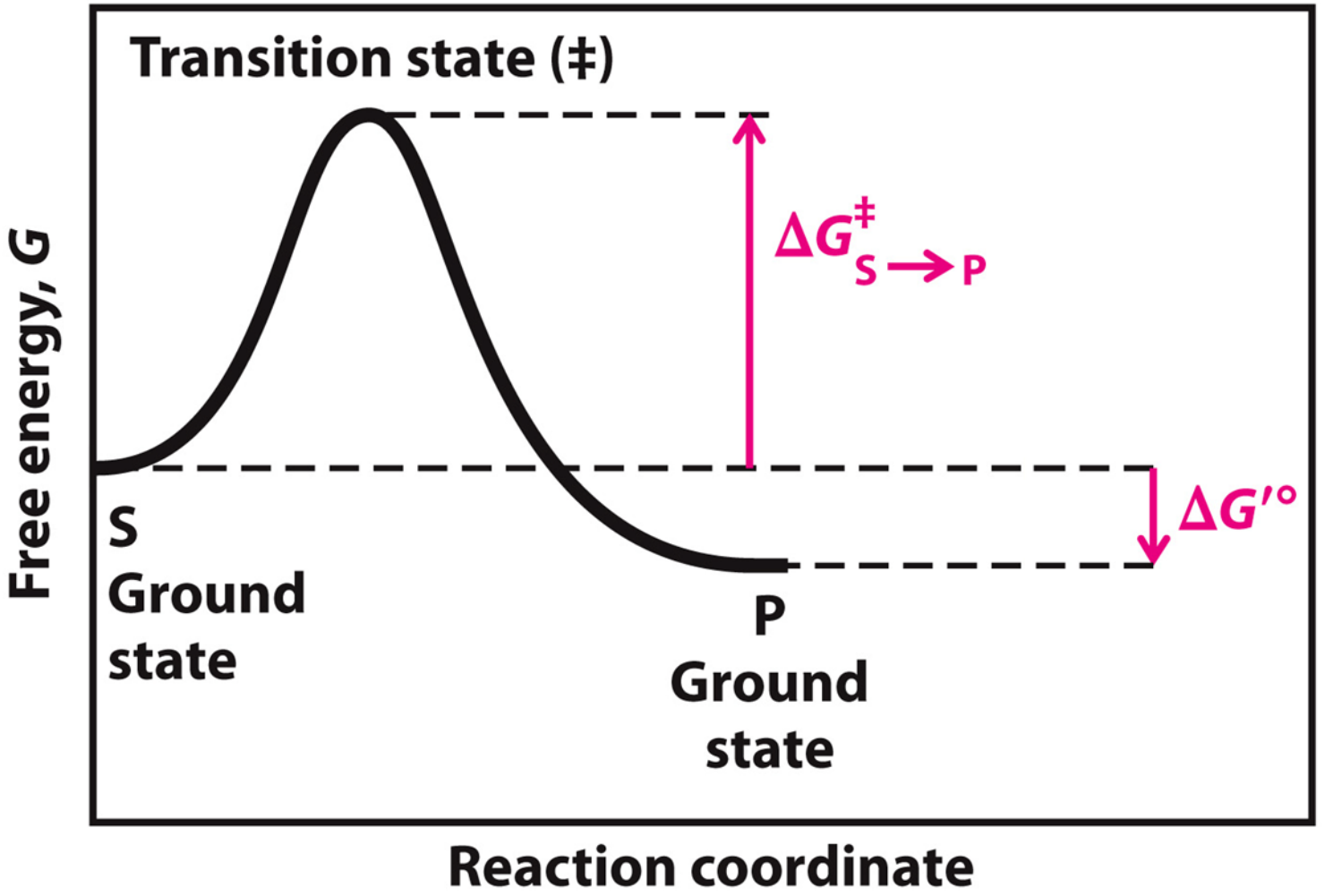


Figure 6-2
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How enzymes work

$S \leftrightarrow P$

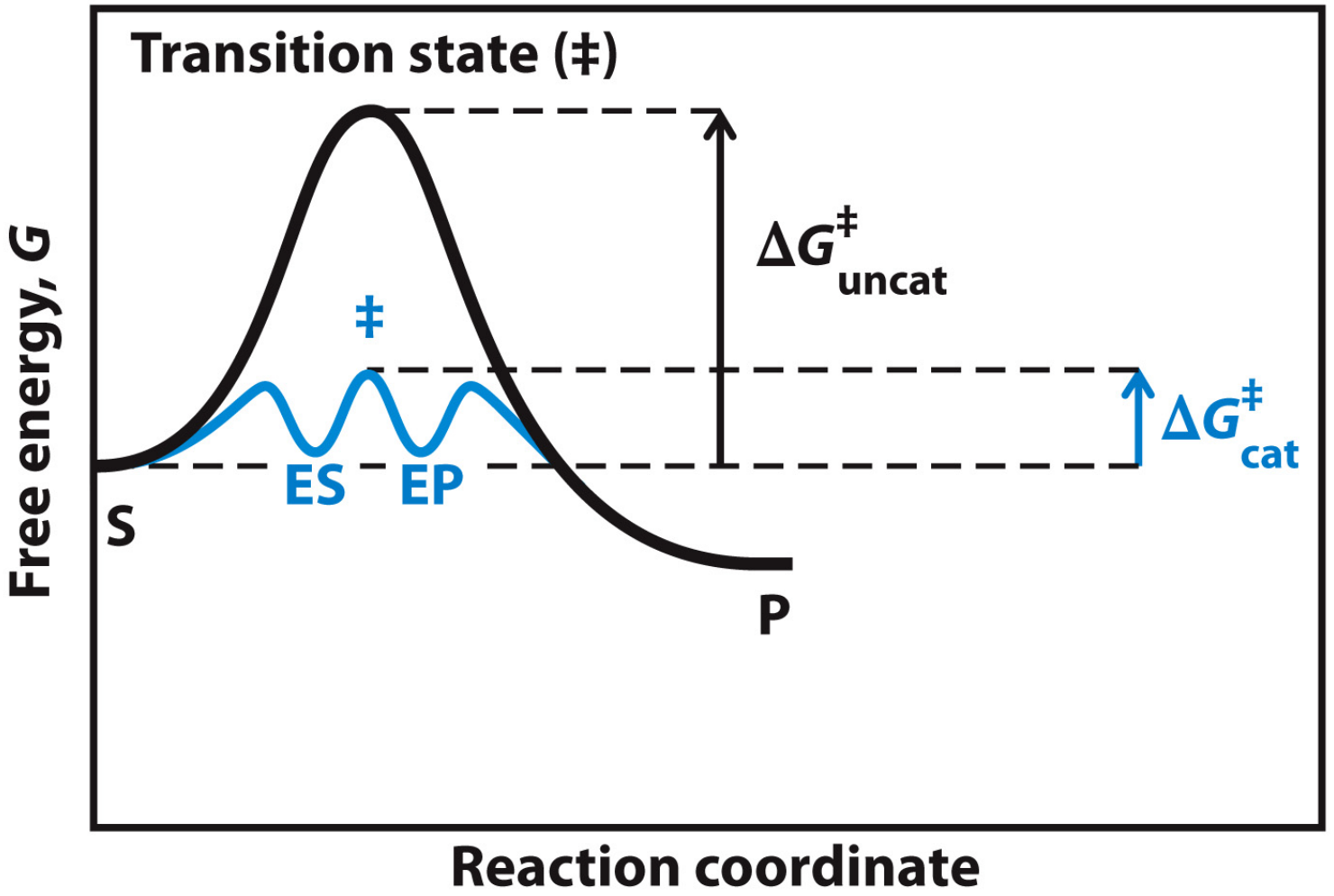


Figure 6-3
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How enzymes work

- Enzymes enhance reaction rates by lowering activation energies
- Enzymes do not affect chemical equilibrium

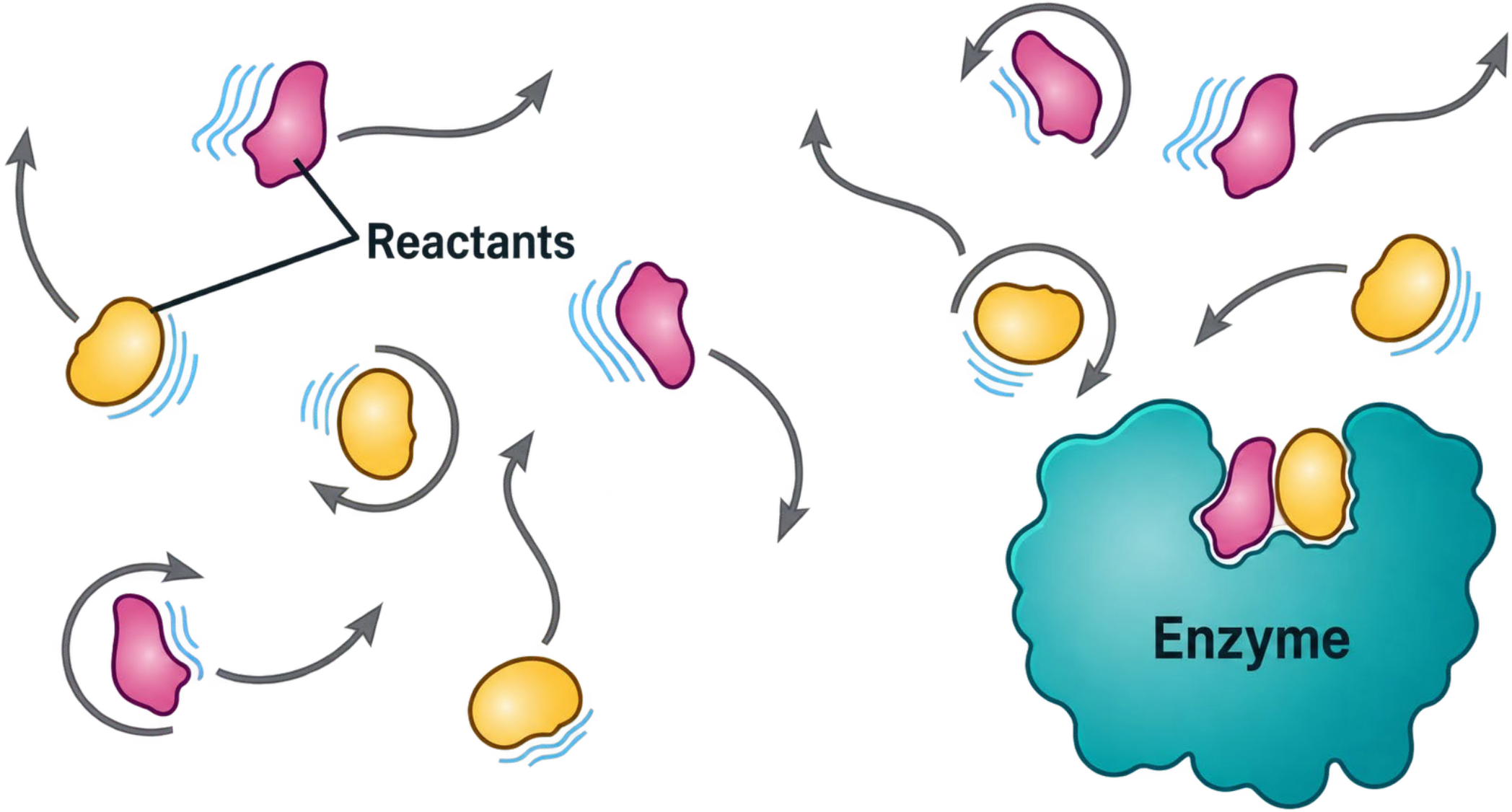
$$\text{Sucrose } (C_{12}H_{22}O_{11}) + 12\ O_2 \rightarrow 12\ CO_2 + 11\ H_2O$$

Mechanisms of enzyme activity

Enzymes lower activation energy through diverse complementary strategies

- Proximity and orientation
- Transition-state stabilization
- General acid-base catalysis
- Covalent catalysis
- Metal ion catalysis …

Effect of proximity and orientation



Transition-state stabilization

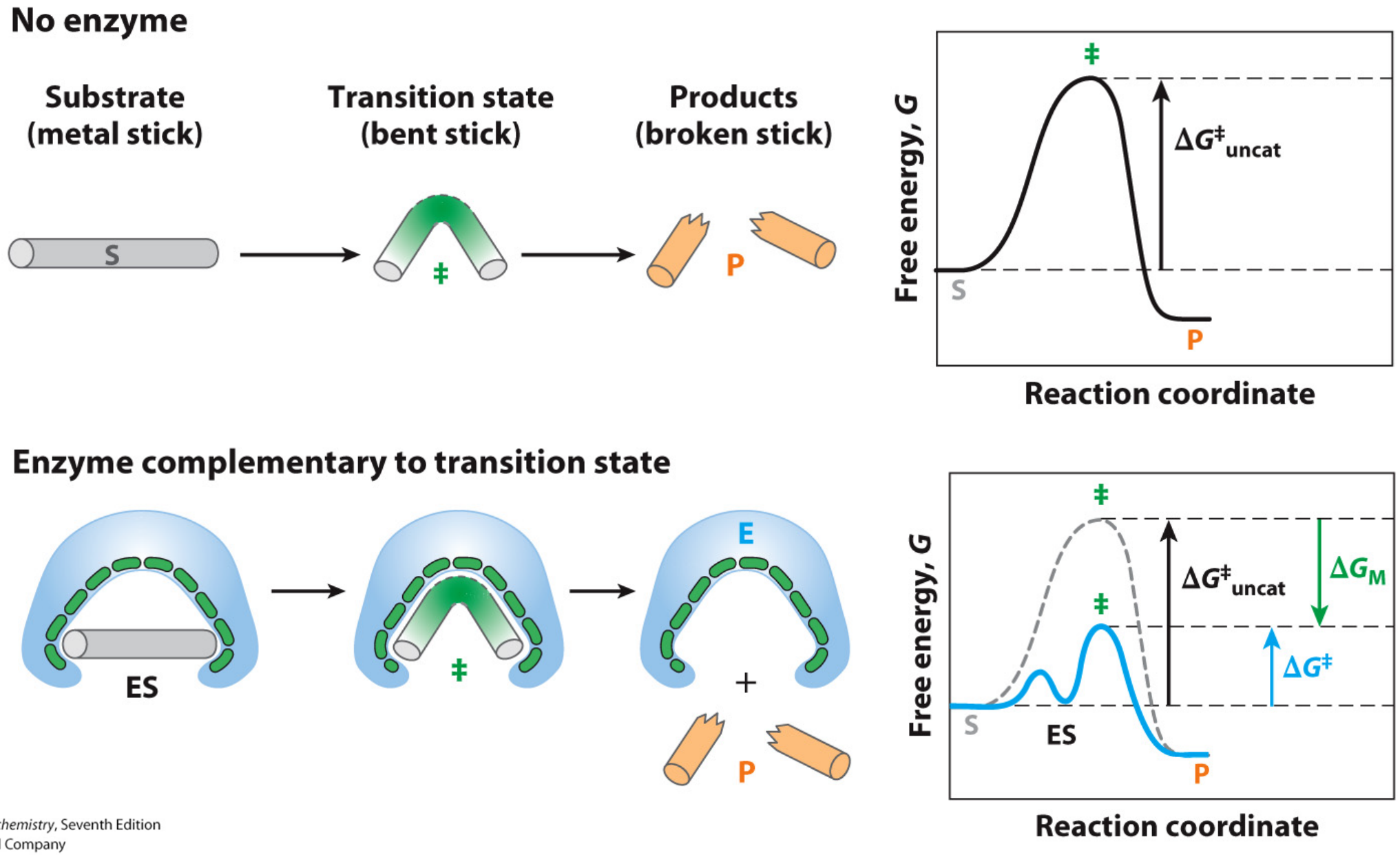
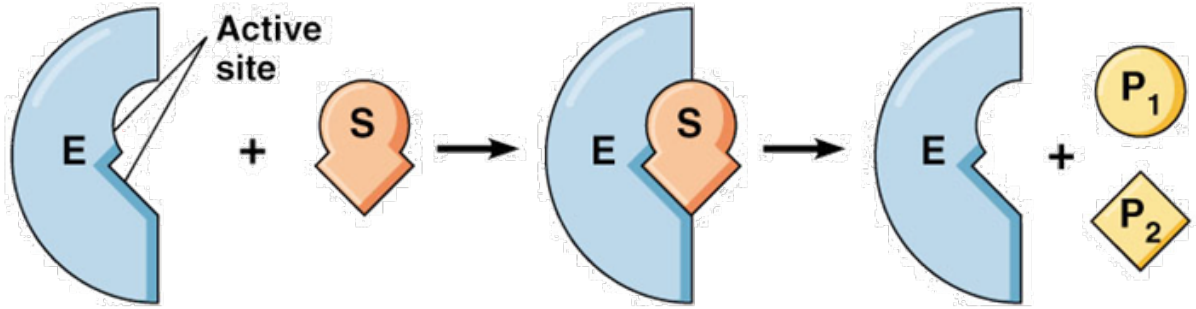


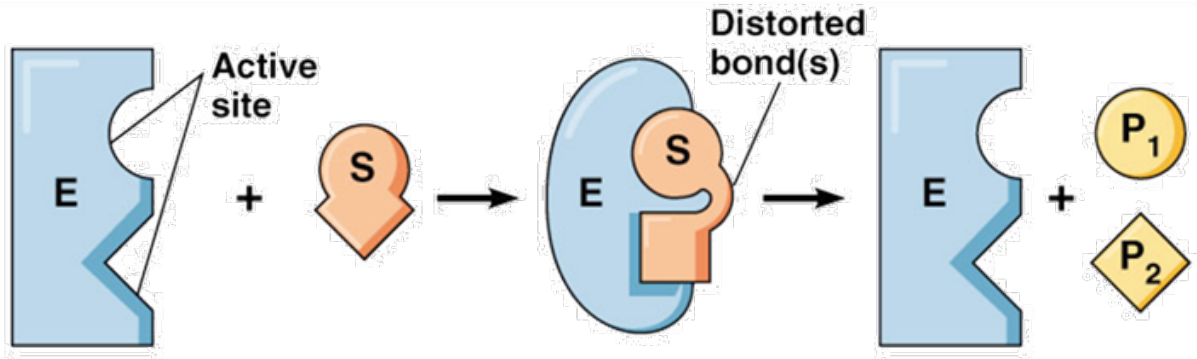
Figure 6-5c
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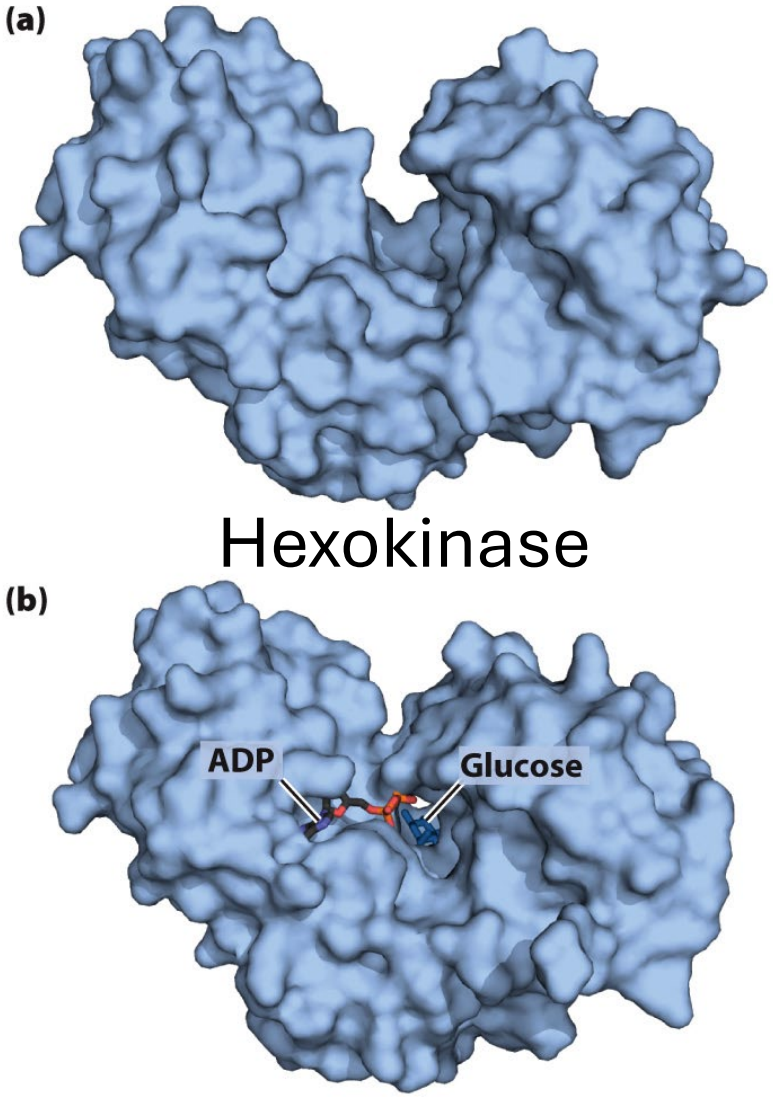
Transition-state stabilization

Fischer's lock-and-key model



Koshland's induced fit model





Hexokinase

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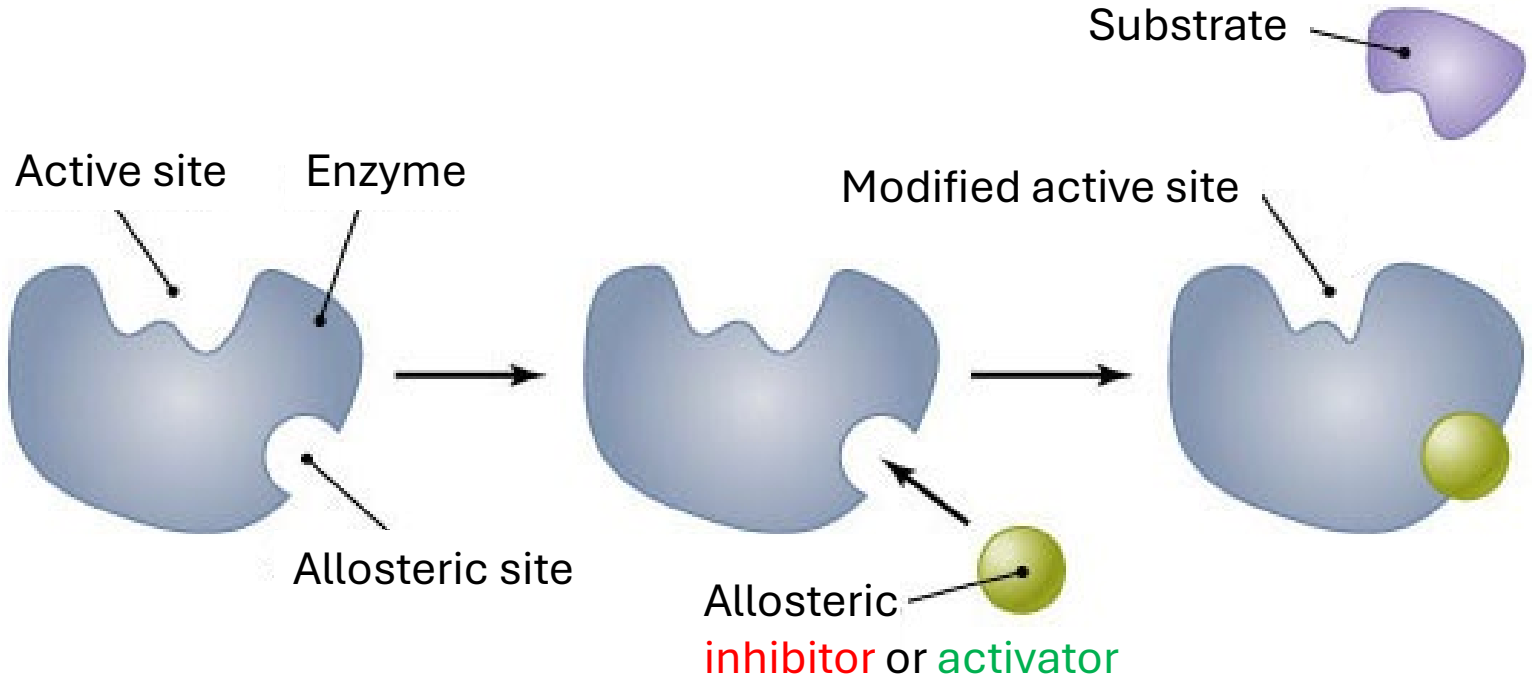
Regulatory enzymes

- Some enzymes exhibit increased or decreased catalytic activity in response to certain signals
- This regulation allows cells to meet changing needs
- Regulation can involve:
  - noncovalent binding of regulatory molecules (allosteric regulators)
  - reversible covalent modification
  - Irreversible proteolytic activation

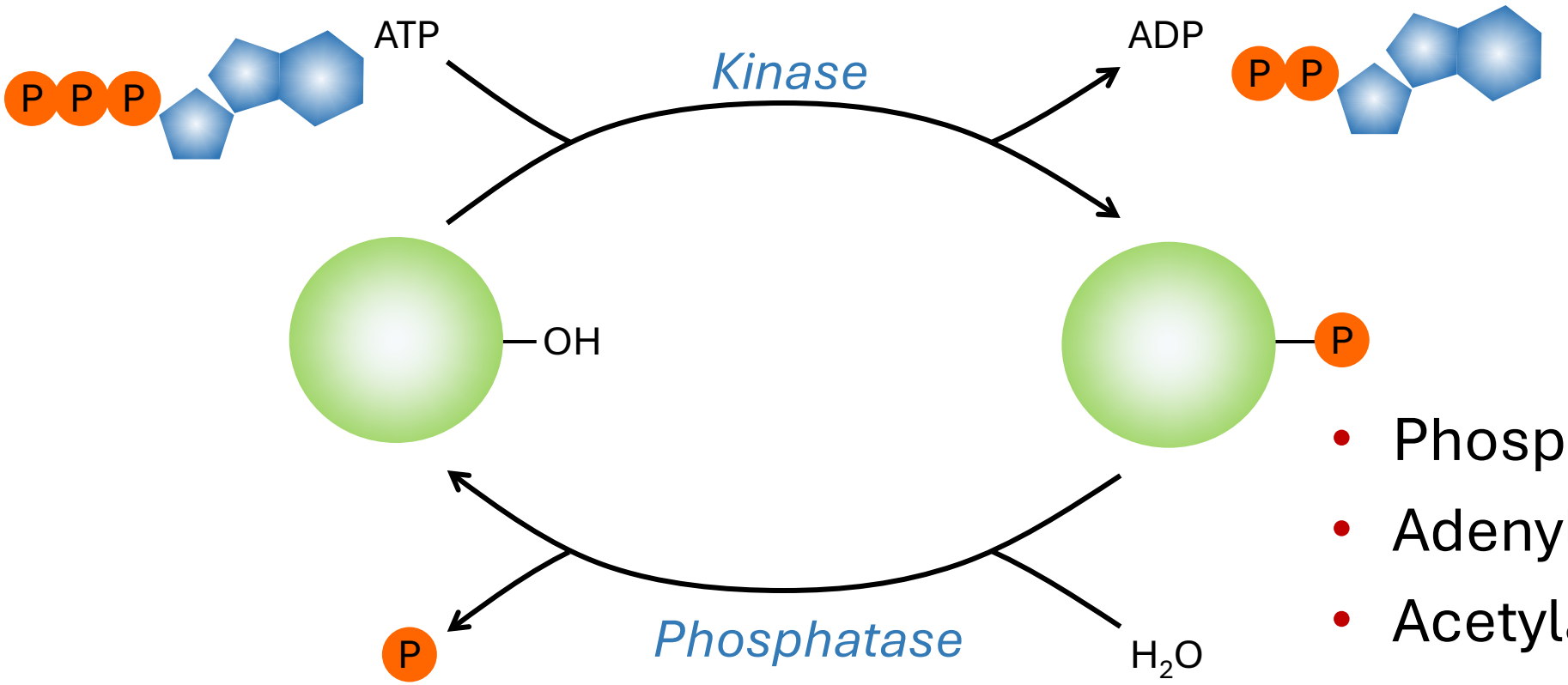
Allosteric regulators

Allosteric effectors or modulators are generally small molecules

There are both allosteric inhibitors and allosteric activators

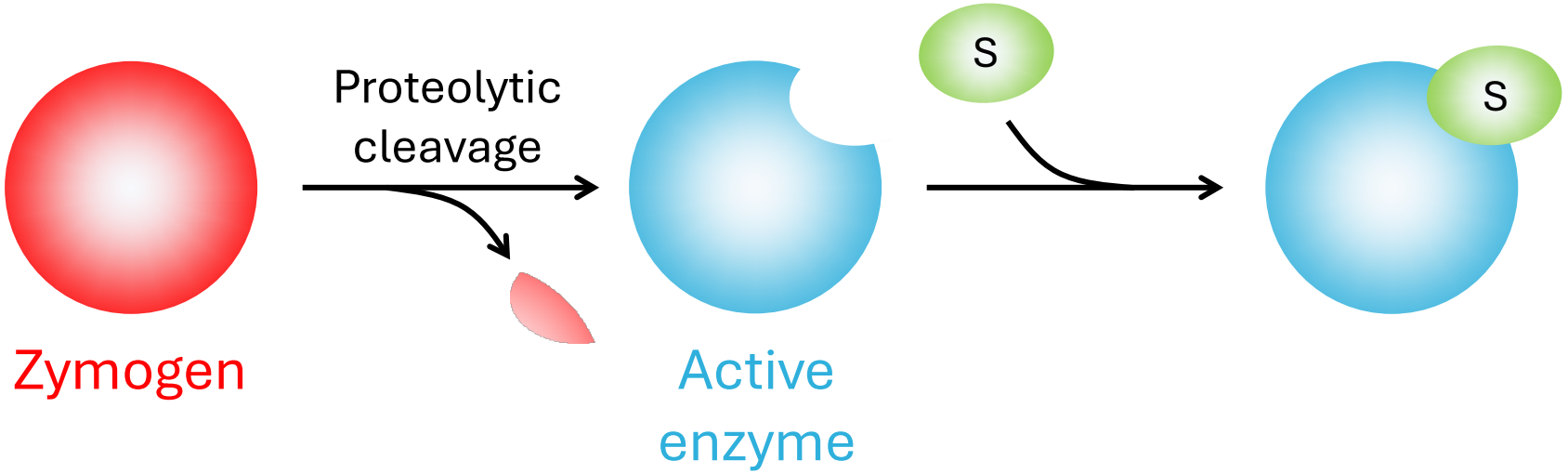


Reversible covalent modification



- Phosphorylation
- Adenylylation
- Acetylation
- Ubiquitination
- ADP-ribosylation
- Methylation …

Irreversible proteolytic activation



- Digestive enzymes
- Coagulation cascade
- Complement system…